

Transcriptómica y Bases de datos especializadas



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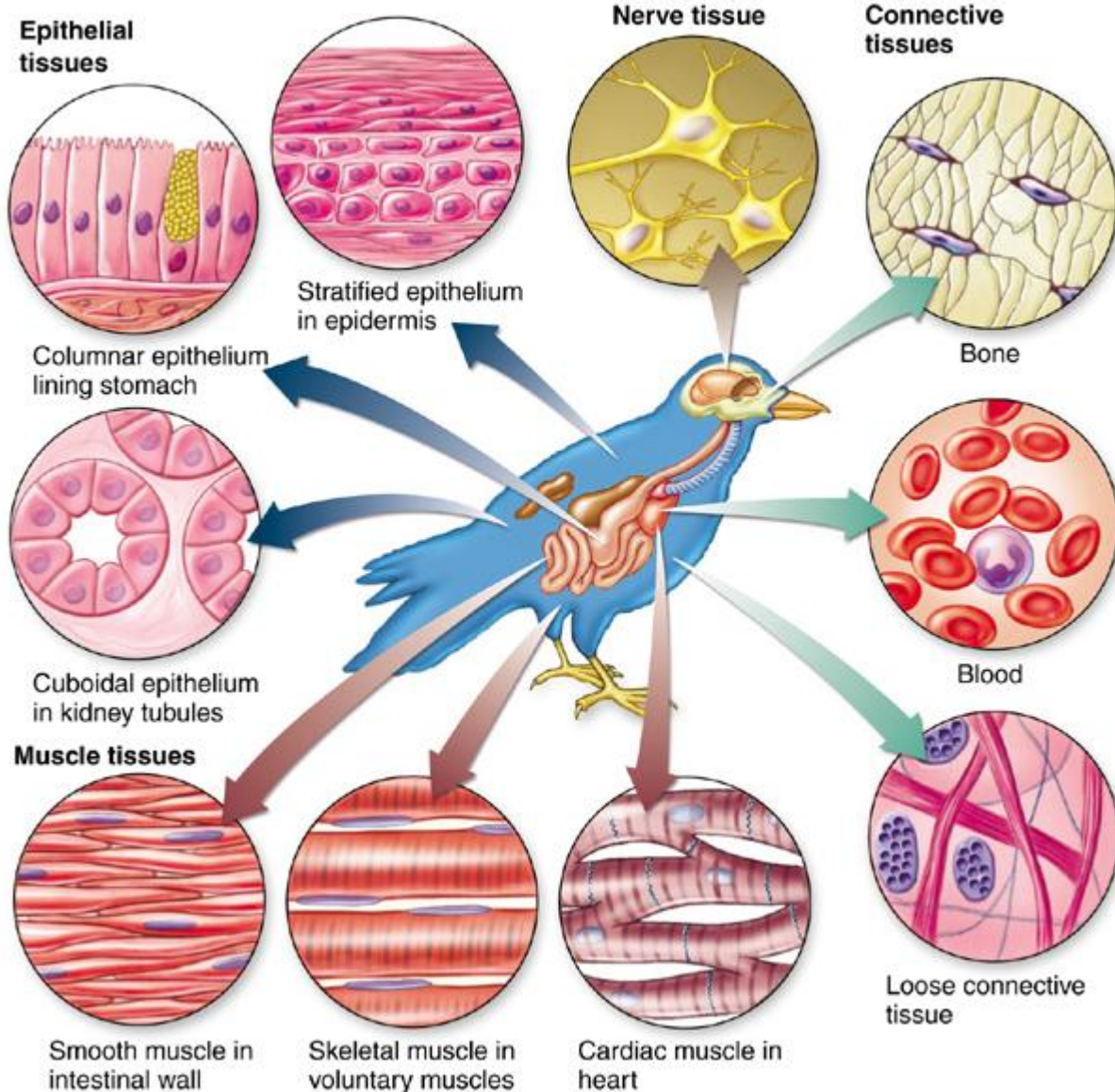
Objetivos de aprendizaje de la clase



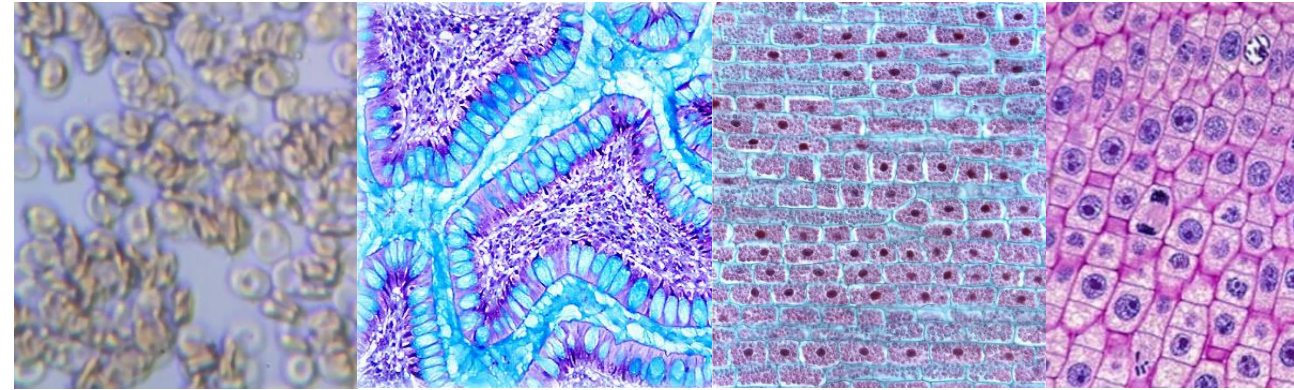
- ☐ Conocer aspectos generales de los transcriptomas
- ☐ Conocer cómo se caracterizan o estudian los transcriptomas
- ☐ Conocer bases de datos especializadas en transcriptómica.

¿Qué hace diferente a una célula de otra?

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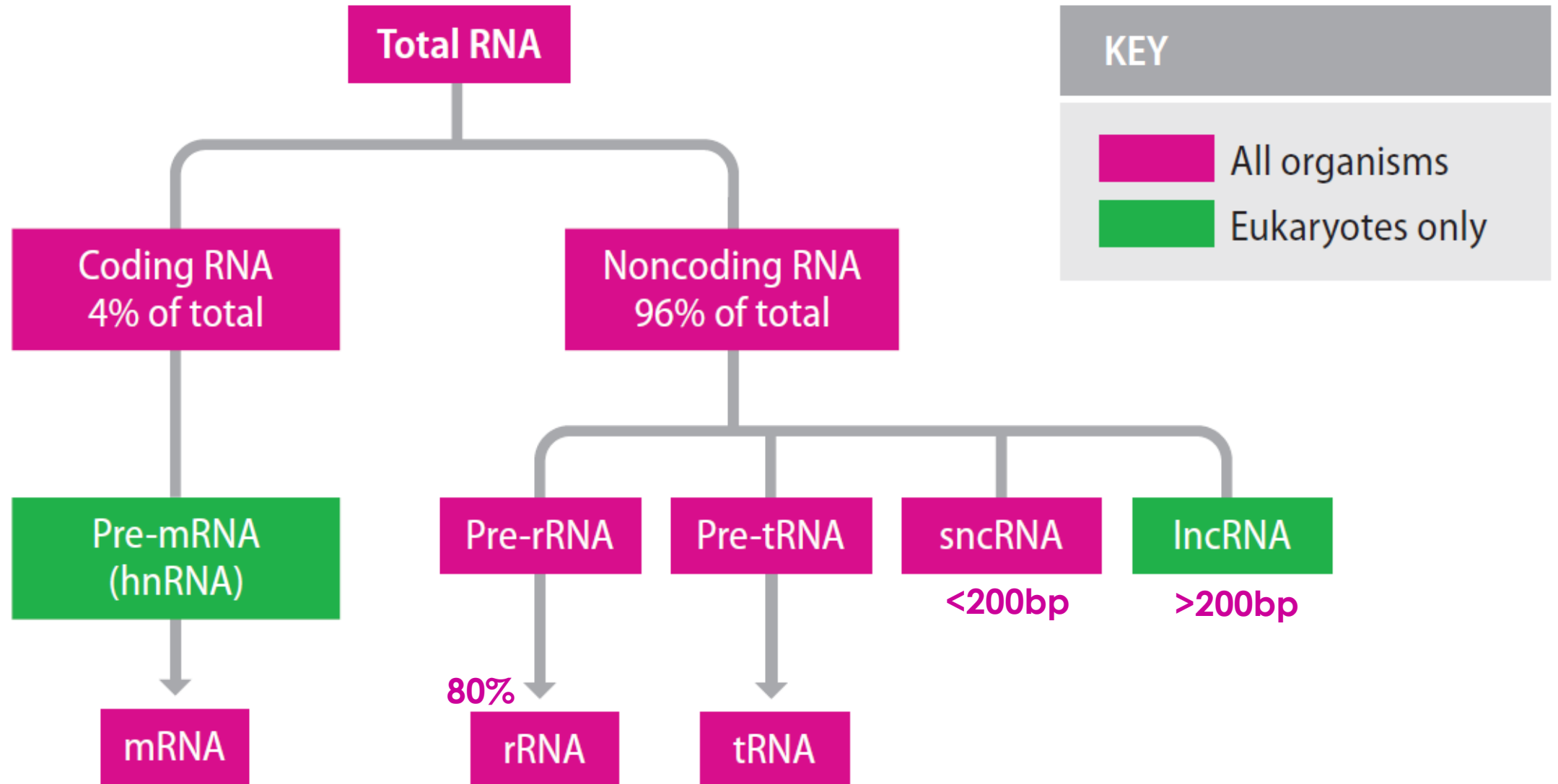


Ya que todas las células contienen la misma información genética (3GB de DNA), ¿Qué hace la diferencia?



Los ~20,000 genes codificantes y ~20,000 no-codificantes de nuestro genoma actúan como ingredientes en una receta que son usados (expresados) por las diferentes células en diferentes tiempos y de diferentes maneras.

¿Qué tipos de RNAs constituyen un transcriptoma?

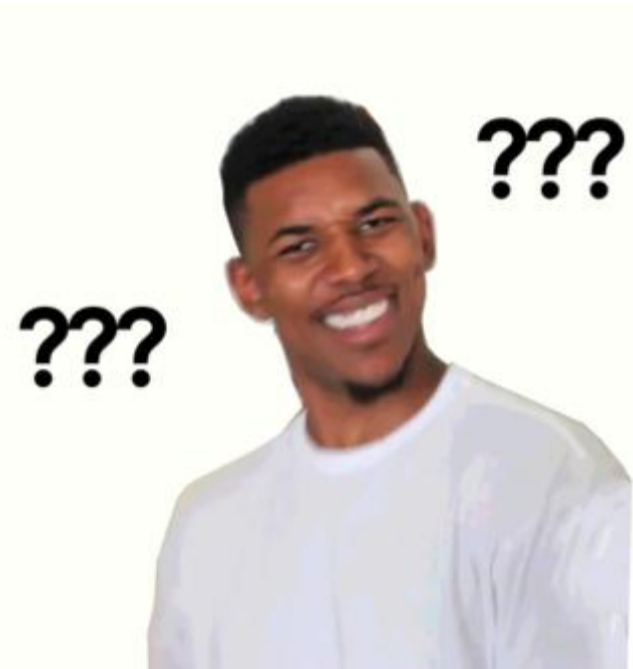


Genomas, transcriptomas y complejidad de los organismos

¿Cuál es la relación entre la talla del transcriptoma y la complejidad de los organismos?

¿Qué nos dicen estudios más recientes acerca de la talla del transcriptoma y su complejidad?

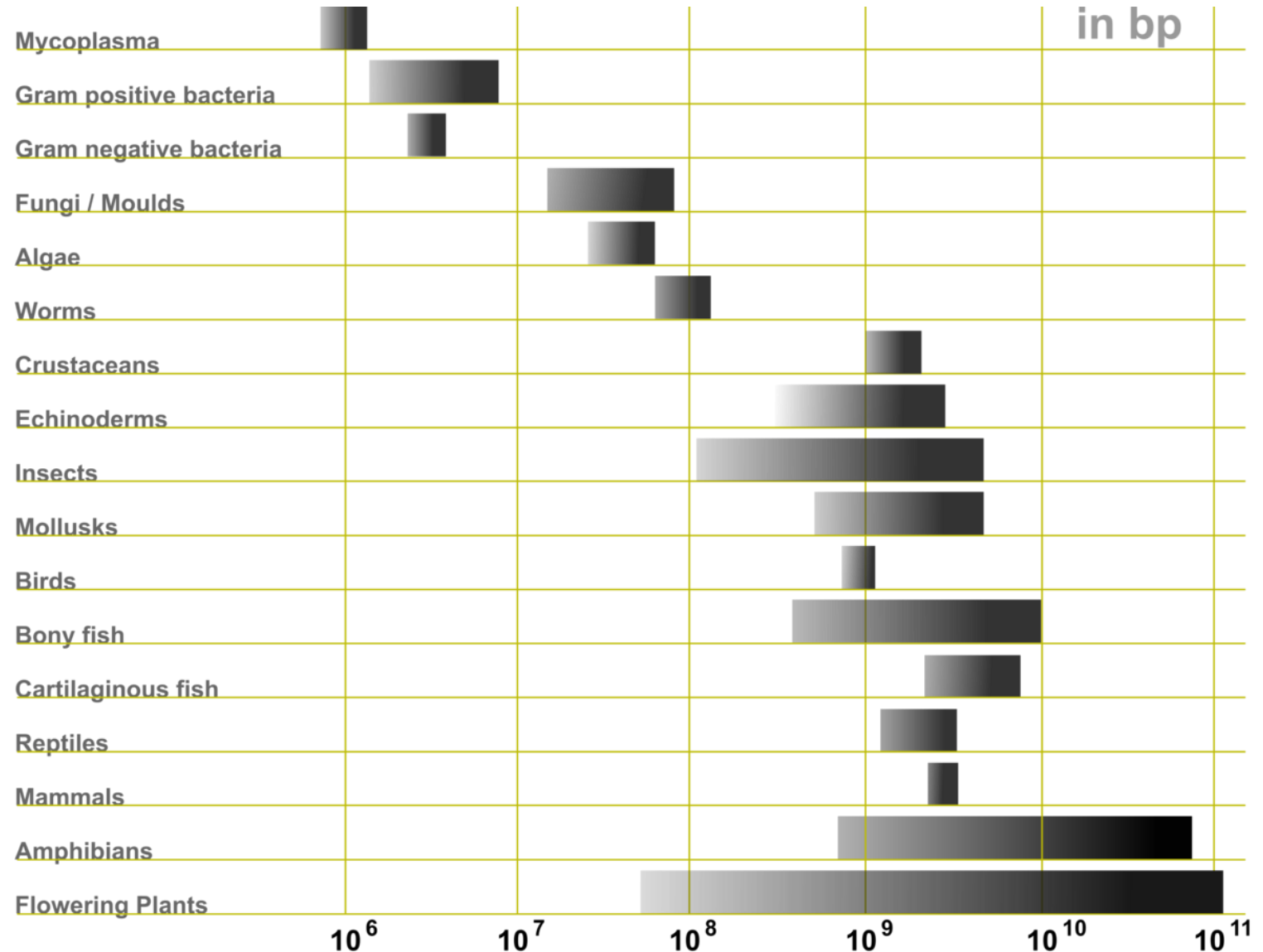
¿Podría la talla del genoma determinar la complejidad de un organismo?



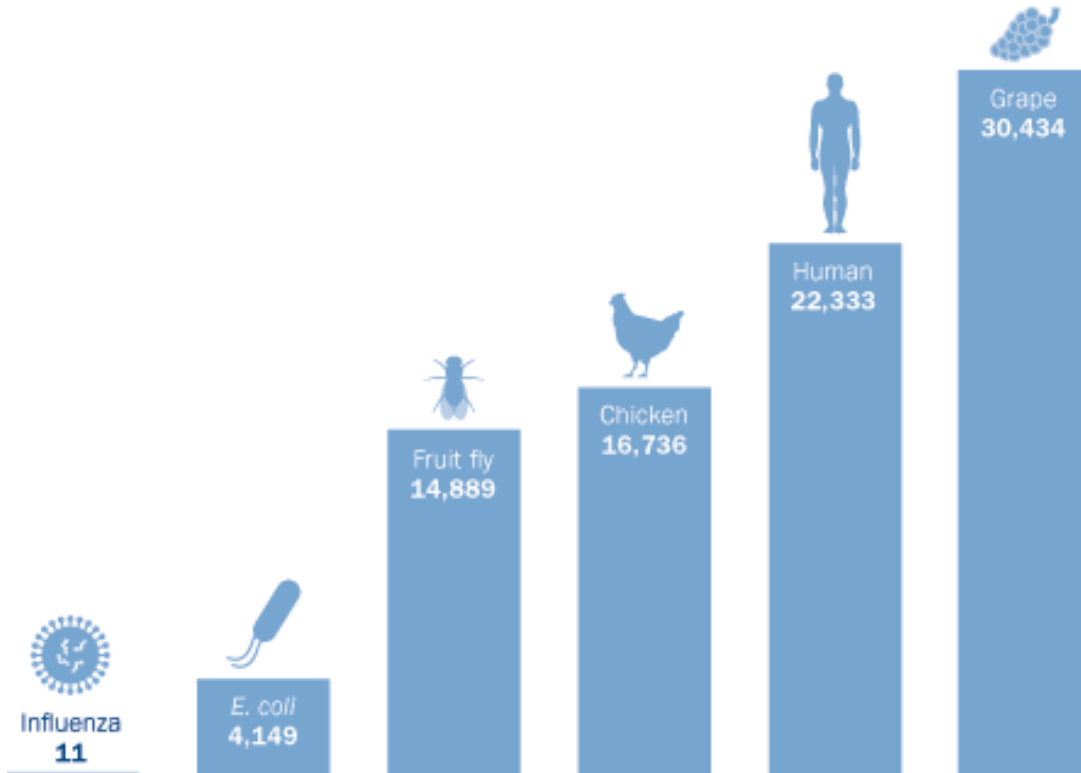
¿Está relacionado el contenido de DNA en un organismo a la complejidad morfológica es este?

No existe una clara correlación entre la talla del genoma y la complejidad genética.

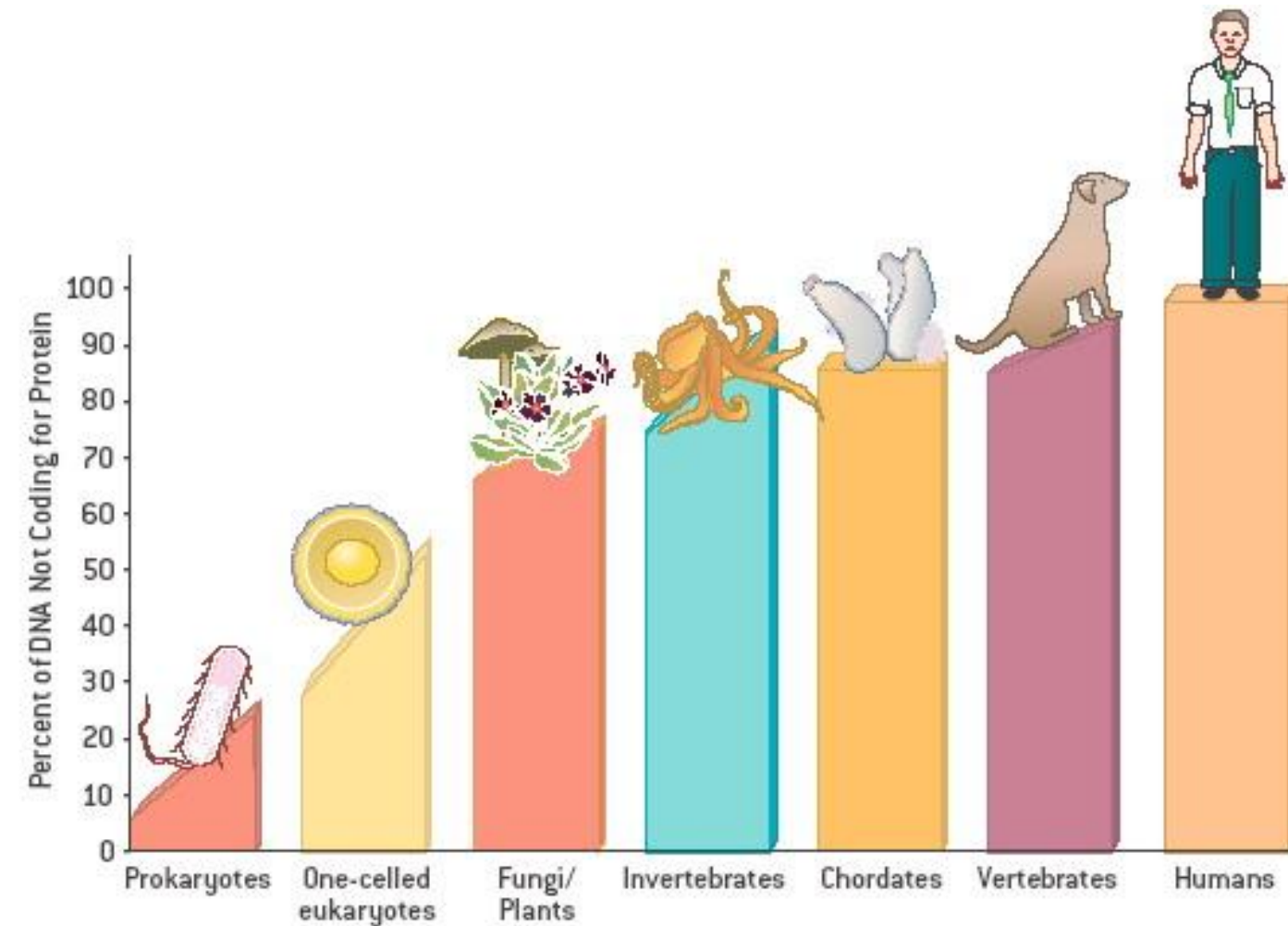
De hecho, organismos muy similares pueden tener tallas genómicas muy distintas (anfibios).



Pero.... y el número de genes codificantes mostrará alguna relación???



Secuencias no-codificantes aumentan con la complejidad del organismo

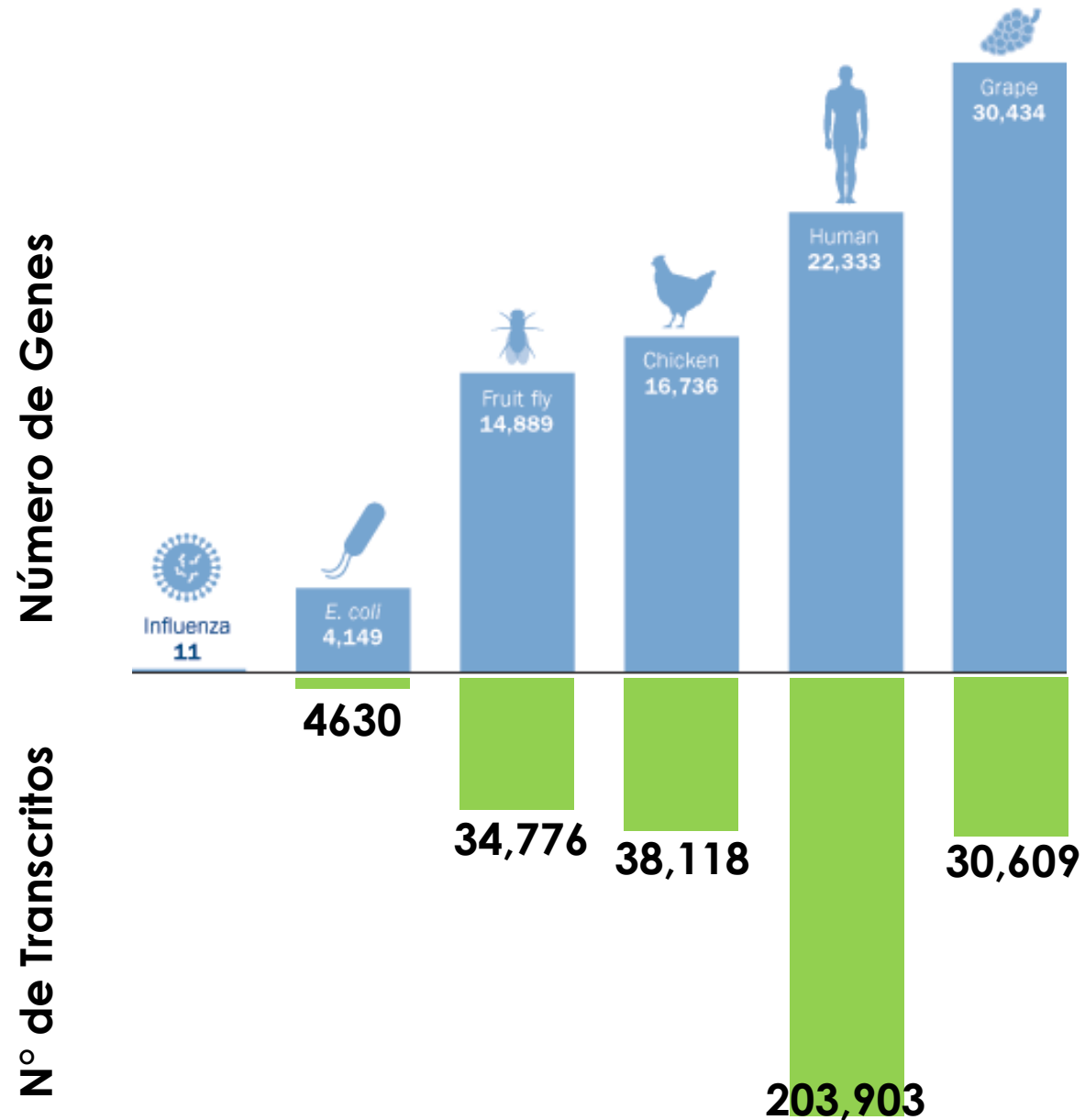


En procariontes, las secuencias no codificantes representan una pequeña porción del genoma.

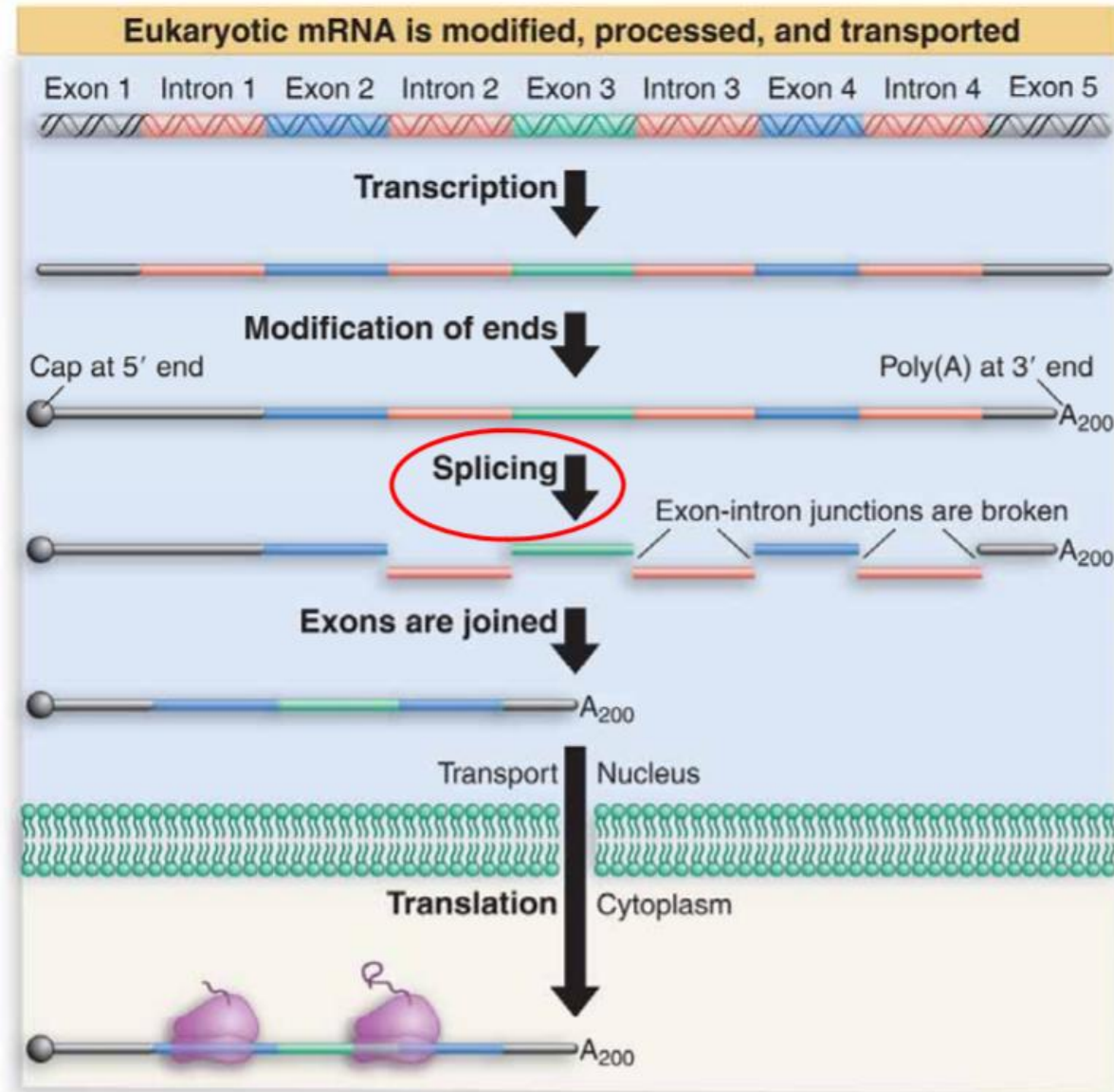
En eucariontes, al aumentar la complejidad, aumenta (por lo general) también la proporción del genoma cubierta por secuencias no codificantes.

Por mucho tiempo las secuencias codificantes fueron consideradas “basura genética”, pero este tipo de secuencias podrían explicar parte de la complejidad de los organismos

Pero.... y el número de transcritos???

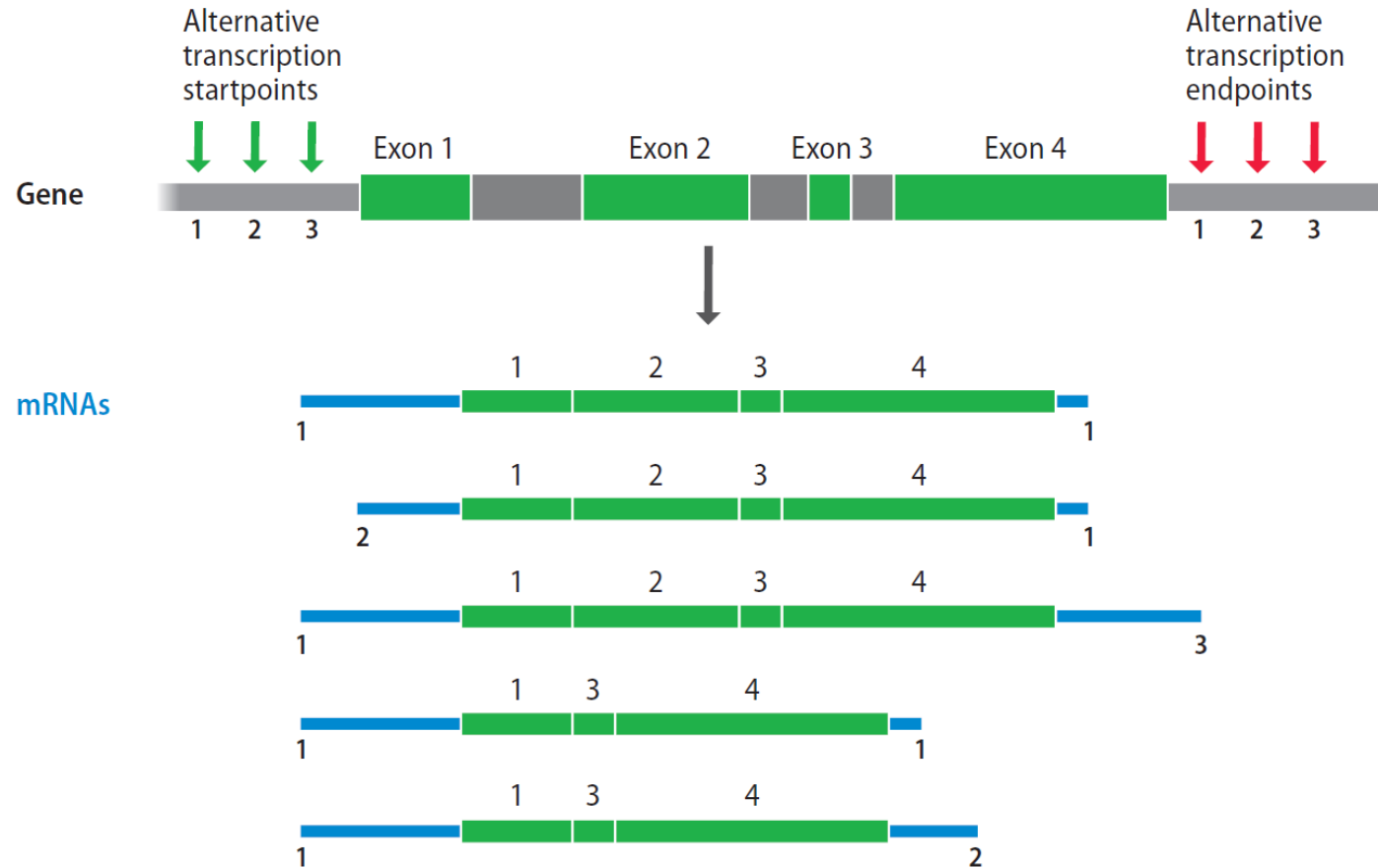


¿Cómo explicamos que ese gran número de transcritos??

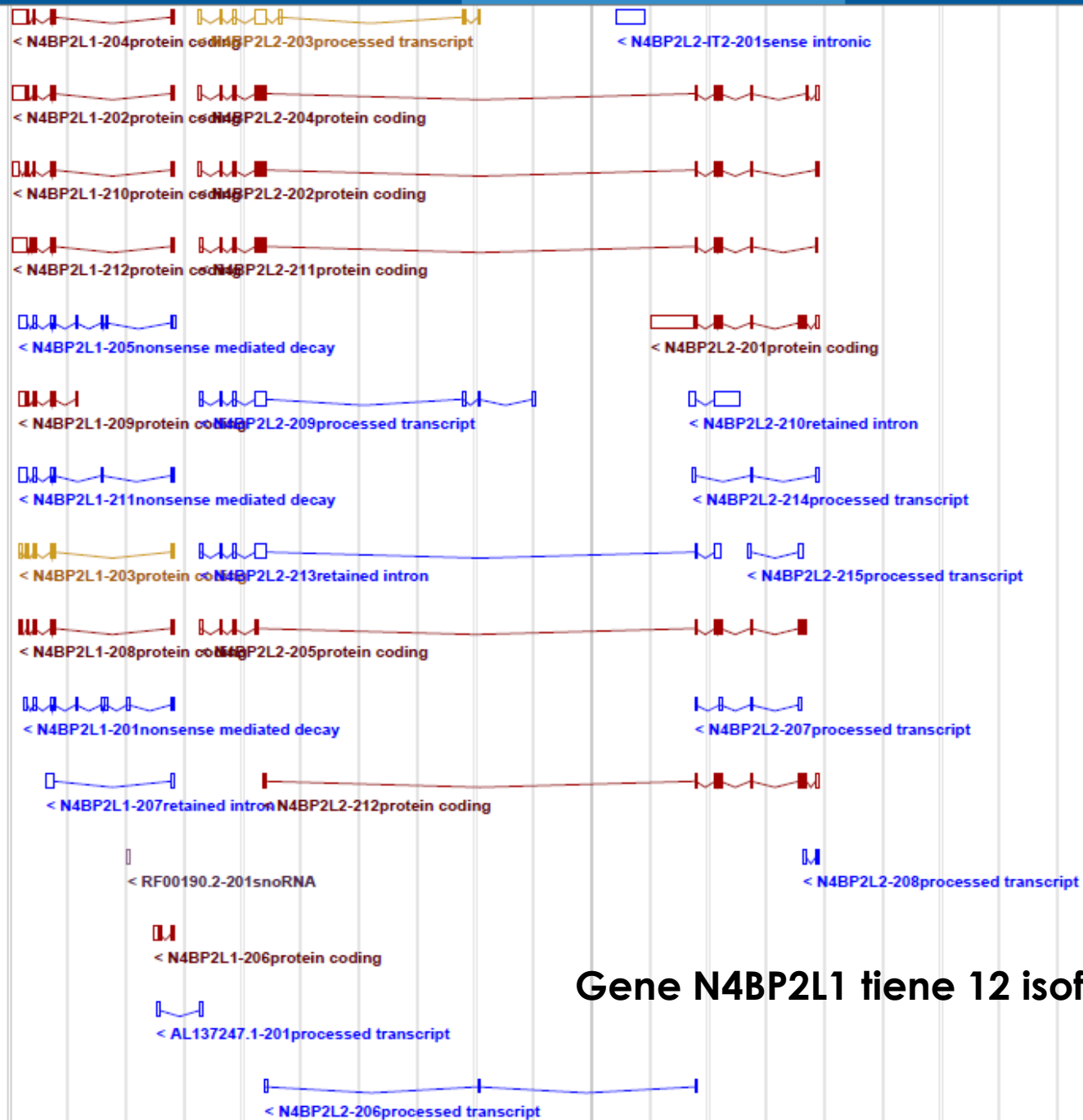


Un gen puede dar origen a muchos mRNAs distintos

El proceso de splicing alternativo aumenta la talla (**complejidad**) del **transcriptoma**



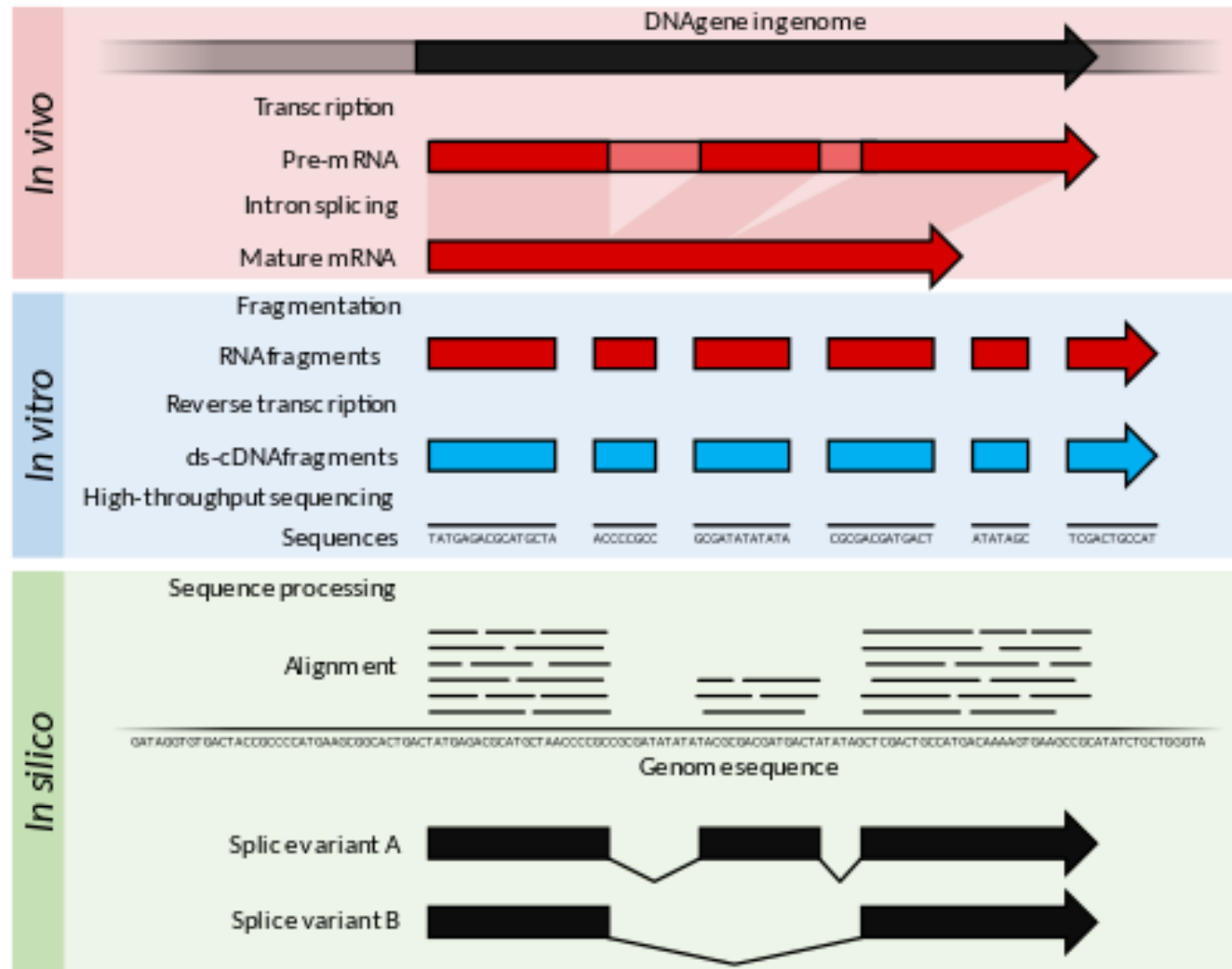
Debido al proceso de splicing y las vías de procesamiento, los ~15,000 genes activos en una célula pueden dar origen a más de 100,000 mRNAs diferentes



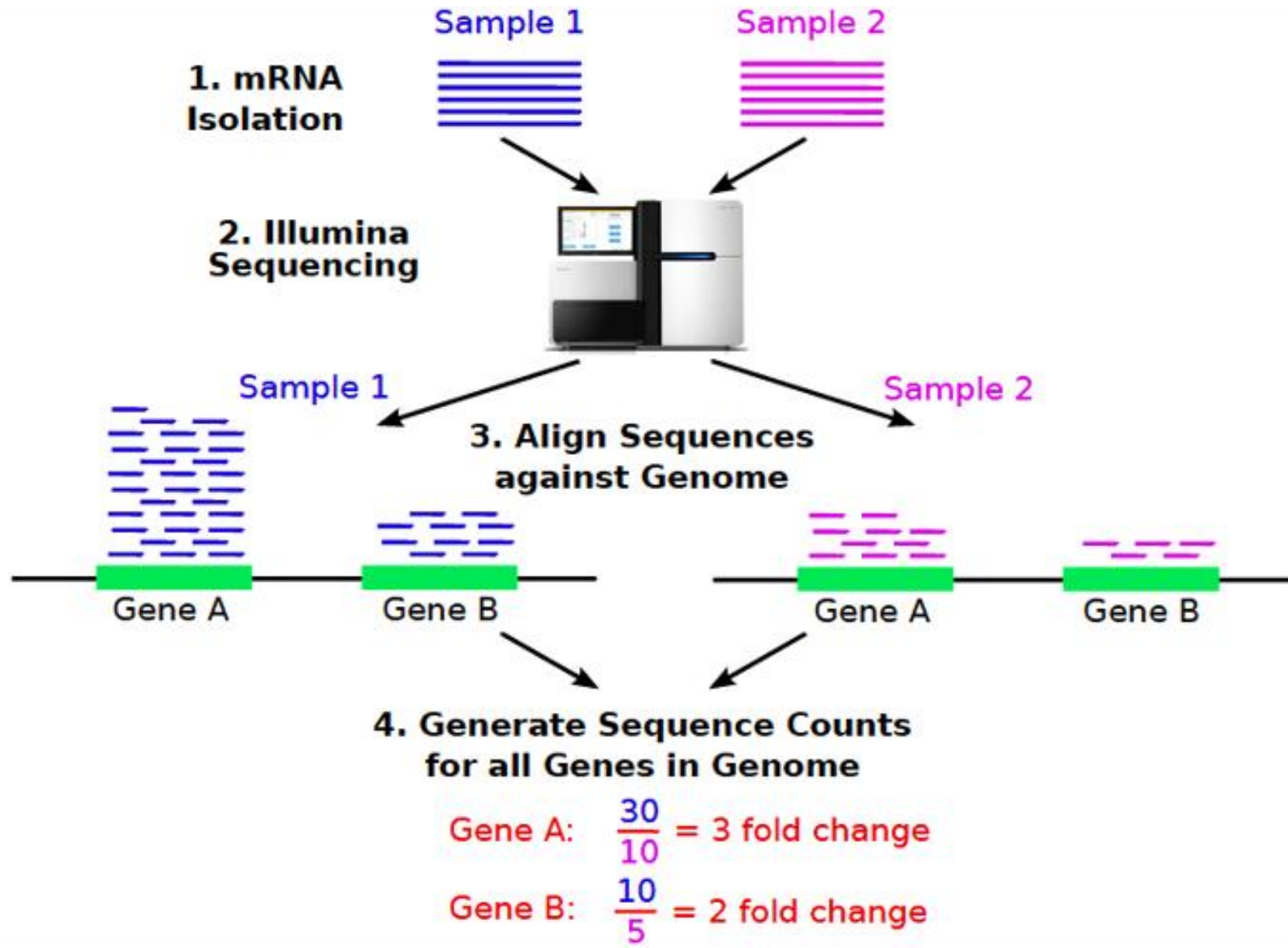
Gene N4BP2L1 tiene 12 isoformas de splicing

¿Cómo caracterizamos el transcriptoma de un organismo?

Aplicando la metodología denominada *RNA-sequencing*



RNA-seq es una técnica cuantitativa



RNA-seq es capaz de detectar y cuantificar todos los tipos de RNA

Diferentes protocolos que se ajustan a las diferentes poblaciones de RNAs

mRNA-Seq

Cuantifica expresión de genes codificantes, identificación de isoformas nuevas y conocidas, fusión de genes, etc.

Total RNA-Seq

Detección de genes codificantes, además de múltiples formas de RNA no-codificantes. Mide abundancia y detecta diversas características de los transcritos.

Targeted RNA-Seq

Selección y secuenciación específica de transcritos de interés, a través de un enriquecimiento o amplificación. Cualitativa y cuantitativa.

Small RNA-Seq

Aislamiento y Secuenciación de Small RNAs tales como microRNA

Single Cell RNA-Seq

Tecnología que permite evaluar el transcriptoma de células individuales

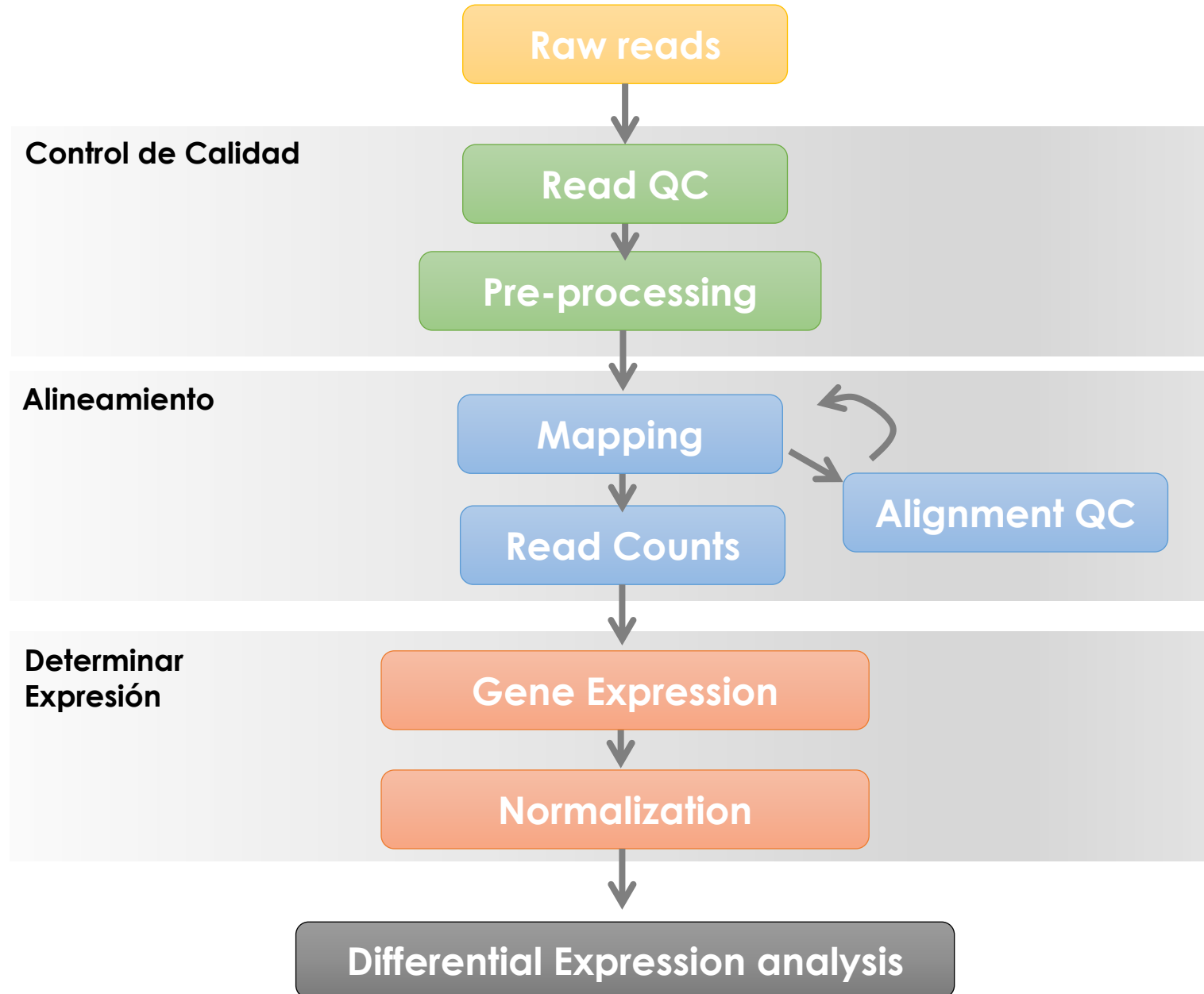
Ribosome Profiling

Secuenciación de fragmentos de mRNA protegidos por ribosomas. Indica que proteínas están siendo traducidas en manera activa.

RNA Exome Capture Sequencing

Captura solo las regiones codificantes del transcriptoma

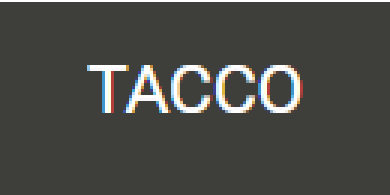
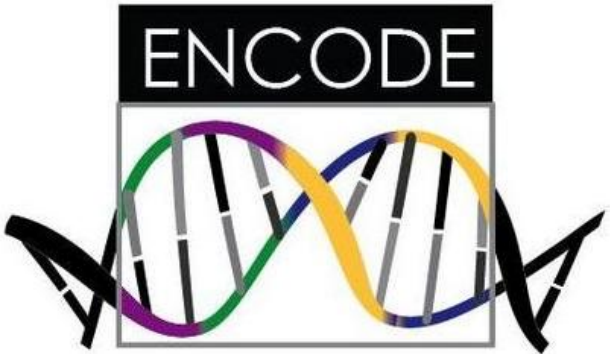
Análisis de datos de RNA-sequencing



Bases de datos y atlas de expresión transcriptómica de disponibilidad pública



National Human Genome Research Institute

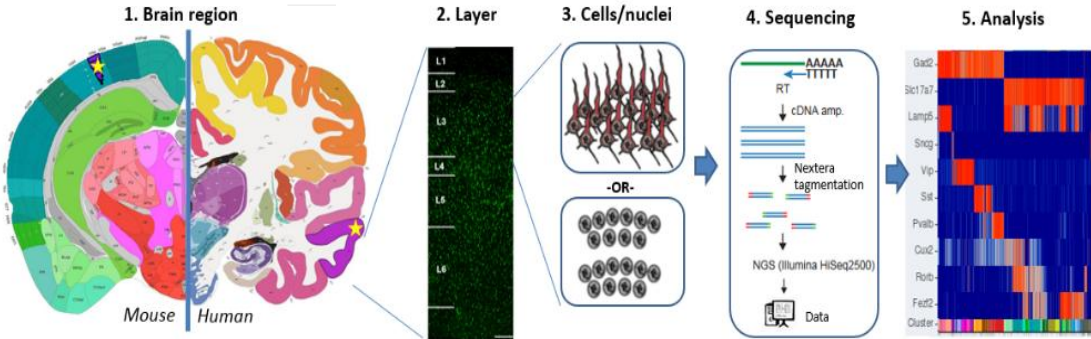


A Cell Atlas of Worm

The *C. elegans* transcriptome at single cell resolution



ALLEN BRAIN ATLAS DATA PORTAL



Bases de datos y Recursos OMICos

Repositorio de datos

Son bases de datos que contienen data de NGS de disponibilidad pública. Contienen datos sin procesar (Raw read), Información del ensamble de las secuencias y anotación funcional. En este tipo de bases de datos es posible descargar datos pero no visualizarlos de manera interactiva

Recursos OMICs

Son bases de datos que contienen datos omicos analizados de manera detalla permitiendo su visualización interactiva. Se caracterizan por ser especializados, por ejemplo, para una especie y/o un tejido determinado

European Nucleotide Archive – Repositorio de datos



Examples: [BN000065](#), [histone](#)

Search

[Advanced](#)

[Sequence](#)

European Nucleotide Archive

The European Nucleotide Archive (ENA) provides a comprehensive record of the world's nucleotide sequencing information, covering raw sequencing data, sequence assembly information and functional annotation. [More about ENA](#)

Access to ENA data is provided through the browser, through search tools, large scale file download and through the API.

Popular

- [Submit and update](#)
- [Sequence submissions](#)
- [Genome assembly submissions](#)
- [Submitting environmental sequences](#)
- [Citing ENA data](#)
- [Rest URLs for data retrieval](#)
- [Rest URLs to search ENA](#)

Text Search

Beta cells zebrafish

Examples: [BN000065](#), [histone](#)

Search

[Advanced search](#)

Latest ENA news

12 Jul 2017: [Submission service maintenance - 14/7/17 to 17/7/17](#)

Webin submission services will not be available between Friday 14/7...

European Nucleotide Archive - Repositorio de datos

Study: PRJEB10137

[Contact Helpdesk](#)

RNAseq from mature ductal cells from nkx6.1:GFP zebrafish lines

View: [Project XML](#) [Study XML](#)

Download: [Project XML](#) [Study XML](#)

Name

Transcriptome of mature ductal cells

Submitting Centre

GIGA-R, University of Liege

Secondary accession(s)

ERP011343

Description

Background: In contrast to mammals, the zebrafish has the remarkable capacity to regenerate very efficiently its pancreatic beta cells. Understanding the mechanisms of regeneration in zebrafish and the differences with mammals will be fundamental to discovering molecules able to stimulate the regeneration process in mammals. To identify the pancreatic cells able to give rise to new beta cells in zebrafish, we generated new transgenic lines allowing the tracing of multipotent pancreatic progenitors and endocrine precursors. Results: Using novel bacterial artificial chromosome transgenic nkx6.1 and ascl1b reporter lines, we established that nkx6.1-positive cells give rise to all the pancreatic cell types and ascl1b-positive cells give rise to all the endocrine cell types in the zebrafish embryo. These two genes are initially co-expressed in the pancreatic primordium and their domains segregate, not as a result of mutual repression, but through the opposite effects of Notch signaling, maintaining nkx6.1 expression while repressing ascl1b in progenitors. In adult zebrafish, nkx6.1 expression persists exclusively in the ductal tree at the tip of which its expression coincides with Notch active signaling in centroacinar/terminal end duct cells. Tracing these cells reveals that they are able to differentiate into other ductal cells and into Insulin-expressing cells in normal – non-diabetic – animals. This capacity of ductal cells to generate endocrine cells is supported by the detection of ascl1b in the nkx6.1:GFP ductal cell transcriptome. This transcriptome also reveals, besides actors of the Notch and Wnt pathways, several novel markers such as id2a. Finally, we show that beta cell ablation in adult zebrafish triggers proliferation of ductal cells and their differentiation into Insulin-expressing endocrine precursors.

ductal and endocrine pancreatic p

Showing results 1 - 3 of 3 results

Study accession	Sample accession	Secondary sample accession	Experiment accession	Run accession	Tax ID	Scientific name	Instrument model	Library layout	FASTQ files (FTP)	FASTQ files (Galaxy)	Submitted files (FTP)	Submitted files (Galaxy)	NCBI SRA file (FTP)	NCBI SRA file (Galaxy)	CRAM Index files (FTP)	CRAM Index files (Galaxy)
PRJEB10137	SAMEA3498332	ERS805481	ERX1054380	ERR977397	7955	Danio rerio	Illumina HiSeq 2000	PAIRED	File 1 File 2	File 1 File 2	Fastq file 1 Fastq file 2	Fastq file 1 Fastq file 2	File 1	File 1		
PRJEB10137	SAMEA3498333	ERS805482	ERX1054381	ERR977398	7955	Danio rerio	Illumina HiSeq 2000	PAIRED	File 1 File 2	File 1 File 2	Fastq file 1 Fastq file 2	Fastq file 1 Fastq file 2	File 1	File 1		
PRJEB10137	SAMEA3498334	ERS805483	ERX1054382	ERR977399	7955	Danio rerio	Illumina HiSeq 2000	PAIRED	File 1 File 2	File 1 File 2	Fastq file 1 Fastq file 2	Fastq file 1 Fastq file 2	File 1	File 1		

[Prev](#) [Next](#) Go to page: [Go](#)

Sequence Read Archive - Repositorio de datos

SRA

SRA

pancreatic cells

[Advanced](#)

Search

[Help](#)

SRA

Sequence Read Archive (SRA) makes biological sequence data available to the research community to enhance reproducibility and allow for new discoveries by comparing data sets. The SRA stores raw sequencing data and alignment information from high-throughput sequencing platforms, including Roche 454 GS System®, Illumina Genome Analyzer®, Applied Biosystems SOLiD System®, Helicos Heliscope®, Complete Genomics®, and Pacific Biosciences SMRT®.

Getting Started

[How to Submit](#)

[Log in to SRA \(for updating and troubleshooting submissions\)](#)

[Log in to Submission Portal \(for submitting sequence data\)](#)

[SRA Documentation](#)

[Download Guide](#)

[SRA Fact Sheet \(.pdf\)](#)

Tools and Software

[Download SRA Toolkit](#)

[SRA Toolkit Documentation](#)

[SRA-BLAST](#)

[SRA Run Browser](#)

[SRA Run Selector](#)

Related Resources

[Submission Portal](#)

[Trace Archive](#)

[dbGaP Home](#)

[BioProject](#)

[BioSample](#)

Sequence Read Archive - Repositorio de datos

NCBI [Resources](#) [How To](#) [etarisal](#) [My NCBI](#) [Sign Out](#)

SRA

SRA

pancreatic cells

Search

Create alert Advanced

Help

Access

Public (5,925)

Source

DNA (354)

RNA (5,568)

Type

exome (2)

Other

aligned data (55)

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Summary 20 per page

Send to: Filters: [Manage Filters](#)

View results as an expanded interactive table using the RunSelector. [Send results to Run selector](#)

Search results

Items: 1 to 20 of 5926

<< First < Prev Page 1 of 297 Next > Last >>

☐ [CRISPR screen of BxPC-3:nolFNg](#)

1. 1 ILLUMINA (Illumina HiSeq 2500) run: 30.8M spots, 1.5G bases, 1,002.1Mb downloads
Accession: SRX3103170

☐ [CRISPR screen of BxPC-3:input](#)

2. 1 ILLUMINA (Illumina HiSeq 2500) run: 86.9M spots, 4.3G bases, 2.8Gb downloads
Accession: SRX3103169

☐ [CRISPR screen of BxPC-3:IFNg](#)

3. 1 ILLUMINA (Illumina HiSeq 2500) run: 27.1M spots, 1.4G bases, 882Mb downloads
Accession: SRX3103168

☐ [Illumina HiSeq 2000 sequencing; RNAseq of primary human gallbladder cells, reprogrammed human gallbladder cells, and human pancreatic beta cells](#)

4. 1 ILLUMINA (Illumina HiSeq 2000) run: 39.4M spots, 4G bases, 2.3Gb downloads
Accession: ERX1943607

☐ [Illumina HiSeq 2500 sequencing; RNAseq of primary human gallbladder cells, reprogrammed human gallbladder cells, and human pancreatic beta cells](#)

5. 1 ILLUMINA (Illumina HiSeq 2500) run: 47.5M spots, 4.8G bases, 3.1Gb downloads
Accession: ERX1943606

☐ [Illumina HiSeq 2500 sequencing; RNAseq of primary human gallbladder cells, reprogrammed human gallbladder cells, and human pancreatic beta cells](#)

6. 1 ILLUMINA (Illumina HiSeq 2500) run: 35.1M spots, 3.5G bases, 2.3Gb downloads
Accession: ERX1943605

Results by taxon

Top Organisms [\[Tree\]](#)

Homo sapiens (3277)

Mus musculus (2615)

Danio rerio (23)

Rattus norvegicus (7)

Escherichia virus M13 (4)

Top Bioprojects

Production ENCODE epigenomic... (6)

Production ENCODE functional... (4)

Search in related databases

Database	Access		all
	public	controlled	
BioSample	1,167		1,167
BioProject	201	2	203
dbGaP		6	6
GEO Datasets	7,395		7,395

Find related data

Database: Select

Find items

Search details

pancreatic[All Fields] AND cells[All Fields]

GTEx Project – Recurso Omico

2020-01-06

New Functional Annotation Track in Locus Browser

We have added a new functional annotation track in the GTEx Locus Browser. This track displays the functional annotation for the variants in view on the Locus Browser.

Functional annotation of variants within transcript regions was performed...

Resource Overview

Current Release (V8)

[Tissue & Sample Statistics](#)
[Tissue Sampling Info \(Anatomogram\)](#)
[Access & Download Data](#)
[Release History](#)
[How to cite GTEx?](#)

The Genotype-Tissue Expression (GTEx) project is an ongoing effort to build a comprehensive public resource to study tissue-specific gene expression and regulation. Samples were collected from 54 non-diseased tissue sites across nearly 1000 individuals, primarily for molecular assays including WGS, WES, and RNA-Seq. Remaining samples are available from the GTEx Biobank. The GTEx Portal provides open access to data including gene expression, QTLs, and histology images.

News and Events

Explore GTEx



Browse

By gene ID

By variant or rs ID

[By Tissue](#)

[Histology Image Viewer](#)

Browse and search all data by gene

Browse and search all data by variant

Browse and search all data by tissue

Browse and search GTEx histology images



Expression

[Multi-Gene Query](#)

[Top 50 Expressed Genes](#)

[Transcript Browser](#)

Browse and search expression by gene and tissue

Visualize the top 50 expressed genes in each tissue

Visualize transcript expression and isoform structures



QTL

[Locus Browser](#)

Visualize QTLs by gene in the Locus Browser

GTex Project

☐ Gene expression for GCK (ENSG00000106633.15)

Data Source: GTEx Analysis Release V8 (dbGaP Accession phs000424.v8.p2)

Data processing and normalization



 SUBSET None Sex
 SCALE Log Linear
 TISSUE SORT ▲ ▼
 MEDIAN SORT ▲ ▼
 OUTLIERS On Off

Top

Gene Expression

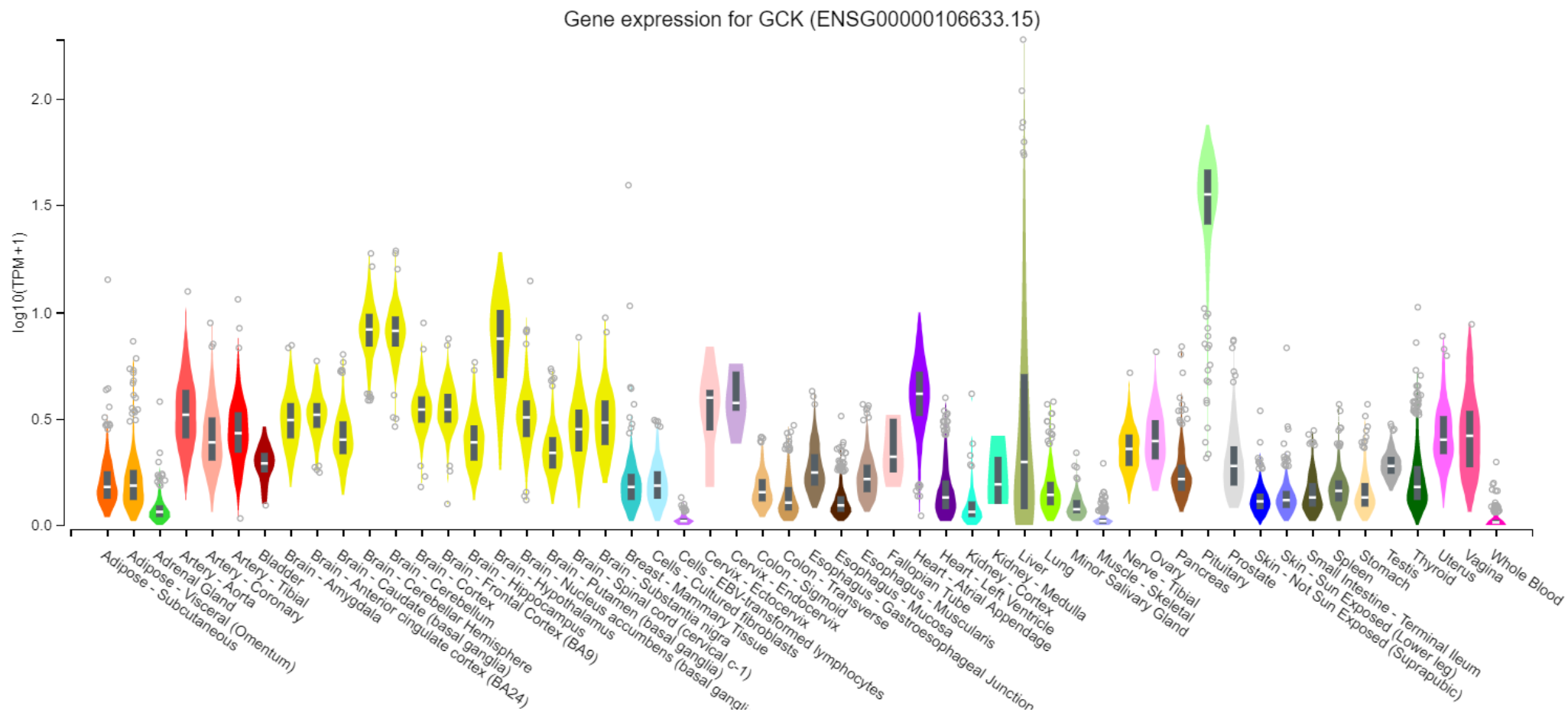
Exon Expression

Single-Tissue eQTLs

Single-Tissue sQTLs

Single-Tissue ieQTLs

Single-Tissue isQTLs



Expression Atlas – Recurso Omico



Expression Atlas

Gene expression across species and biological conditions

Query single cell expression

To Single Cell Expression Atlas →

- Home
- Browse experiments
- Download
- Release notes
- FAQ
- Help
- Licence
- About
- Feedback



Exploring gene expression results across species under different biological conditions

Expression Atlas is an open science resource that gives users a powerful way to find information about gene and protein expression across species and biological conditions such as different tissues, cell types, developmental stages and diseases among others. Expression Atlas aims to help answering questions such as 'where is a certain gene expressed?' or 'how does its expression change in a disease?'.

Read more about Expression Atlas

Search Gene set enrichment

Gene / Gene properties

Enter gene query...

Examples: REG1B, zinc finger, Q14777 (UniProt), GO:0010468 (regulation of gene expression)

Search

Clear

Organism

Any

Biological conditions

Enter condition query...

Examples: lung, leaf, valproic acid, cancer

By species Animals Plants Fungi



Homo sapiens

1200 experiments

29 1171



Mus musculus

976 experiments

40 936



Arabidopsis thaliana

533 experiments

6 527



Drosophila melanogaster

126 experiments

1 127



Saccharomyces cerevisiae

41 experiments

0 41

Rattus norvegicus

141 experiments

2 139

Featured Latest experiments



Human tissues
Human cells



Plasma cells of tonsil
Rare types of haemopoietic cells
Common types of haemopoietic cells



Mouse cells
Mouse tissues
Human tissues



Human tissues



Cancer Cell Line Encyclopedia



Proteomics – Cell lines
RNA – Cell lines



Human tissues



PCAWG by disease
PCAWG by individual



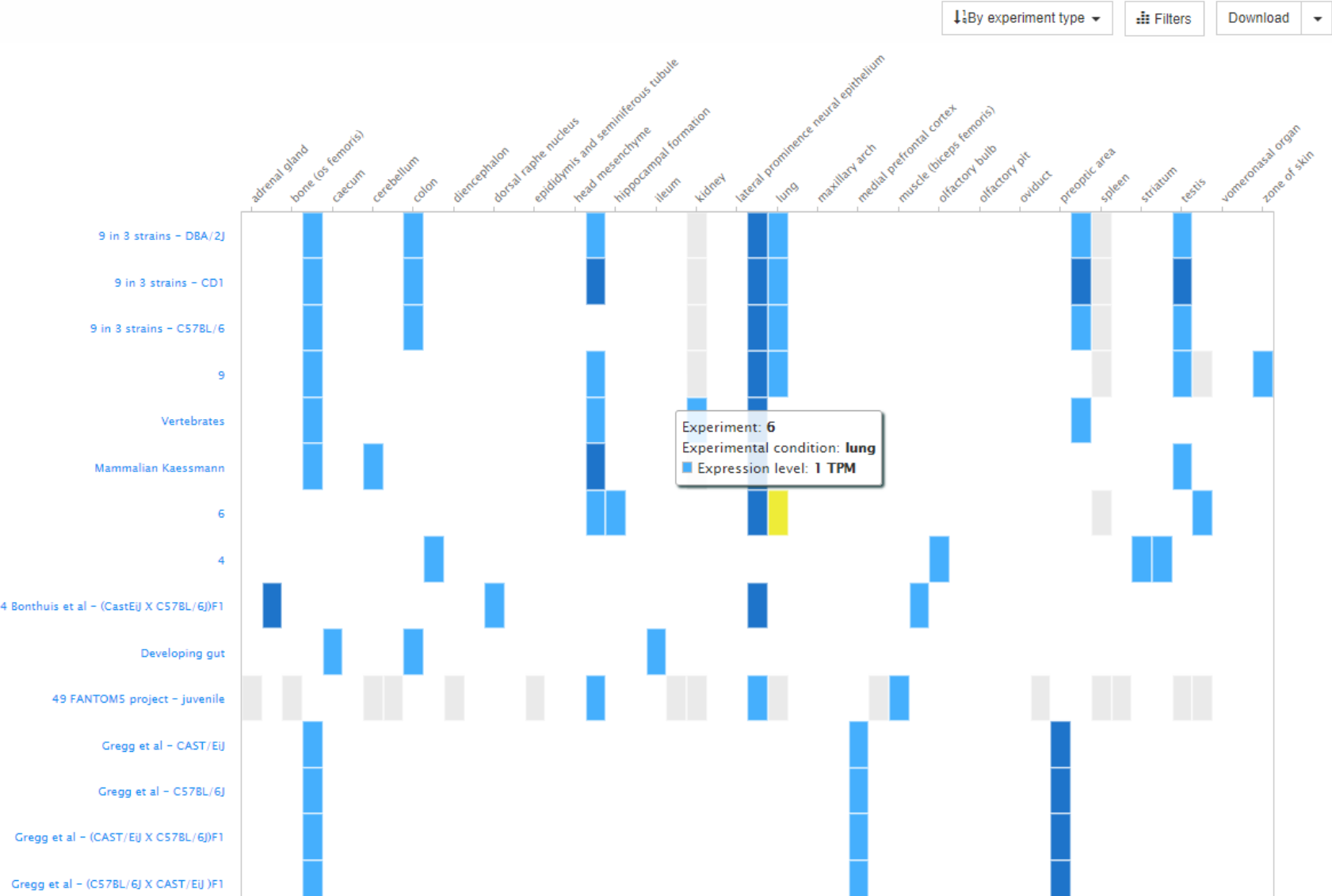
Genomics of Drug Sensitivity in Cancer Project – Cell lines

Expression Atlas

ression Gck information

Organism part

Showing 25 experiments:



TACCO - Transcriptome Alterations in CanCer Omnibus

TACCO

Home

Browse ▾

Select DEGs

Pathway analysis ▾

Upload gene list ▾

Tutorial

Gene Name

GCK ▾

Gene expressions in different cancer types

Correlation with miRNA

Copy

CSV

Excel

PDF

Print

Search:

Cancer type ▴▾	Fold change ▴▾	log2 (Fold change) ▴▾	Mean TPM* (tumor) ▴▾	Mean TPM* (normal) ▴▾	p-value ▴▾
Breast invasive carcinoma	-2.13	-1.09	4.07	8.68	2.33e-34
Pan-kidney cohort (KICH+KIRC+KIRP)	-1.39	-0.47	6.79	9.4	2.04e-24
Colon adenocarcinoma	-2.2	-1.14	3.05	6.7	2.91e-14
Kidney renal clear cell carcinoma	-1.02	-0.03	7.8	7.94	1.57e-13
Thyroid carcinoma	-1.54	-0.62	14.1	21.69	2.61e-12
Uterine Corpus Endometrial Carcinoma	-1.44	-0.53	25.82	37.16	3.43e-12
Prostate adenocarcinoma	-2.1	-1.07	6.03	12.65	6.96e-12
Stomach and Esophageal carcinoma	-1.34	-0.42	7.34	9.83	1.32e-9

HBT – Human Brain Transcriptome



- [Home](#)
- [Gene Search](#)
- [Developmental Trajectories](#)
- [Data](#)
- [Publications](#)
- [Contacts](#)

Links:
[Sestan Lab](#)
[Department of Neurobiology](#)
[School of Medicine](#)
[Kavli Institute](#)
[Yale University](#)

Gene search provides **basic** and **advanced** searches. Basic search provides dynamic gene expression along entire development and adulthood in the cerebellar cortex (CBC), mediodorsal nucleus of the thalamus (MD), striatum (STR), amygdala (AMY), hippocampus (HIP) and 11 areas of neocortex (NCX). Advanced search provides lists of the top 10-100 genes enriched in one of the brain regions.

Basic search

Gene identifier:

Enter gene names (Separated by space):

Brain structure:

Advanced search

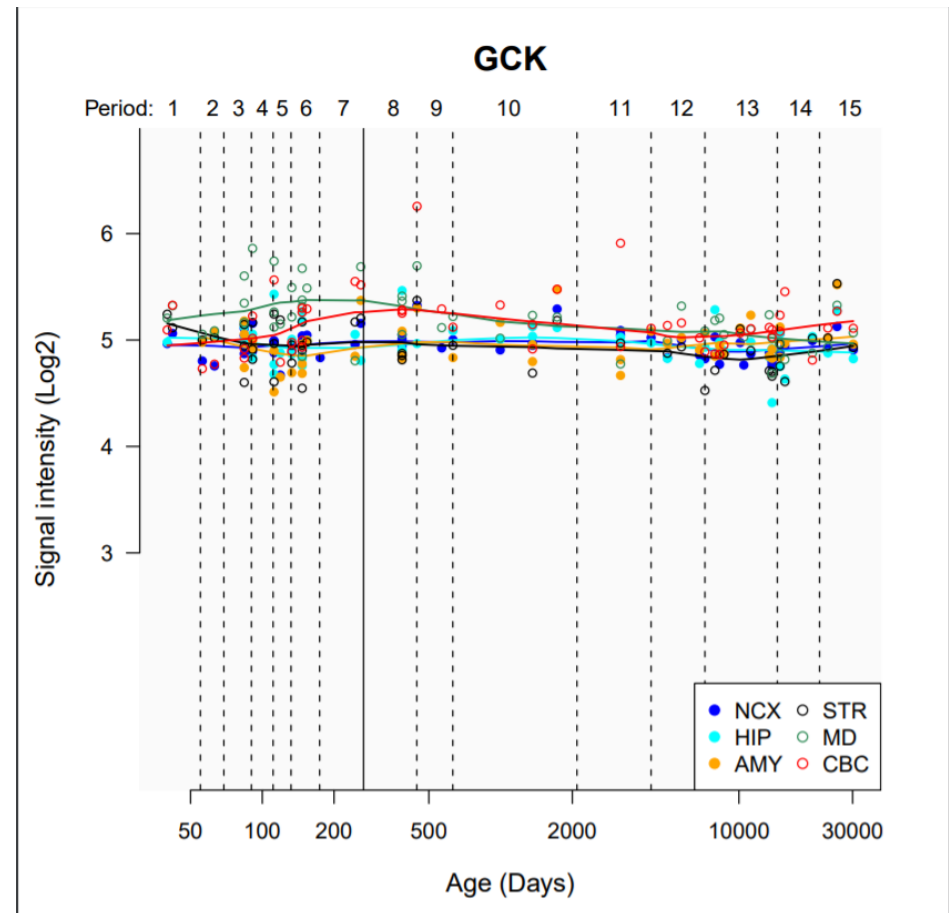
Temporal start period:

Temporal end period:

Brain structure:

Number of top genes:

Please send questions and comments to mingfeng.li@yale.edu



En resumen...

- ❑ La transcriptómica estudia el **set completo de transcritos** (RNAs) expresados en una célula o tejido
- ❑ El transcriptoma está compuesto por **diferentes tipos de RNAs**, donde los mRNA representan 4%
- ❑ La diversidad de los transcriptomas es enriquecida por eventos de **Splicing alternativo** de los genes activos de una célula o tejido
- ❑ La **diversidad** de transcritos en un organismo aumenta con su **complejidad**
- ❑ La secuenciación de RNA (RNA-seq) permite caracterizar los transcriptomas. Diversas **estrategias metodológicas** permitirán estudiar diferentes tipos de RNAs.
- ❑ **RNA-seq** es una técnica **cualitativa y cuantitativa**.
- ❑ Repositorios y recursos omicos, son bases de datos que permiten **acceder a datos de NGS** (procesados o no).